



Lecture 4: Storage and flexibility

Energy Economics of the Green Transition

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1. From costs to value: what generation costs, and what it is worth
2. Four ways to balance a system
3. Storage technologies
4. A workshop on Zerrahn et al. (2018): how much storage does a renewable system need?

From costs to value

Four kinds of cost

- **Investment**, EUR per kW: building the plant. Paid up front, proportional to its size, and *sunk*.
- **Fixed operation and maintenance**, EUR per kW and year: keeping it available, whether or not it runs.
- **Variable**, EUR per MWh: fuel, carbon and wear,

$$c = \frac{p^F}{\eta} + o \quad \text{and} \quad e = \frac{\varphi}{\eta} \text{ t CO}_2 \text{ per MWh.}$$

The marginal cost: it sets the merit order.

- **Adjustment**: what it costs to *change* output. Cold starts, ramps, minimum load. A cost of a pattern, not a level.

Generation Technologies, §2. The first three are lecture 3's model; the fourth is today's.

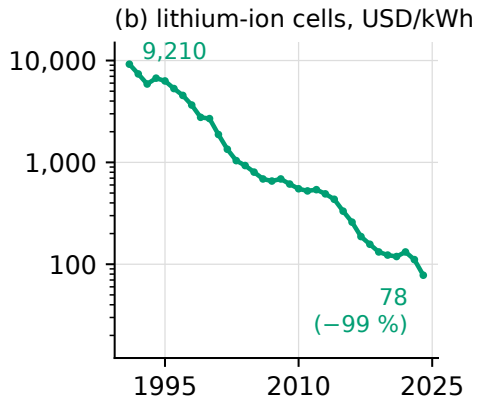
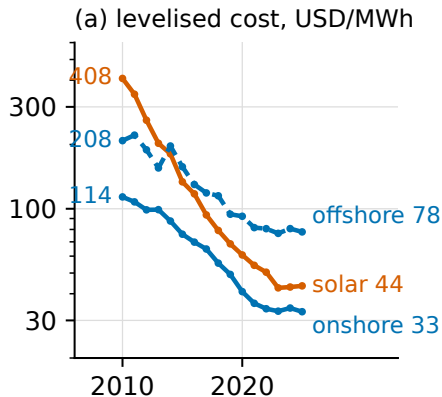
Technologies by cost structure

Technology	Investment	Running	Adjustment	Time to build
Solar PV, utility	Medium	Low	High	Low
Solar PV, rooftop	High	Low	High	Low
Onshore wind	Medium	Low	High	Medium
Offshore wind	High	Low	High	High
Hydro, reservoir	High	Low	Low	High
Hydro, run-of-river	High	Low	High	High
Nuclear	Very high	Low	High	Very high
Coal	High	Medium	High	High
Gas CCGT	Medium	Medium	Medium	Medium
Gas OCGT (peaker)	Low	High	Low	Low
Oil peaker	Low	Very high	Low	Low
Biomass CHP	High	Medium	High	Medium
Waste CHP	Very high	High	High	Medium

Generation Technologies, Table 1. Levels are relative to the other rows.

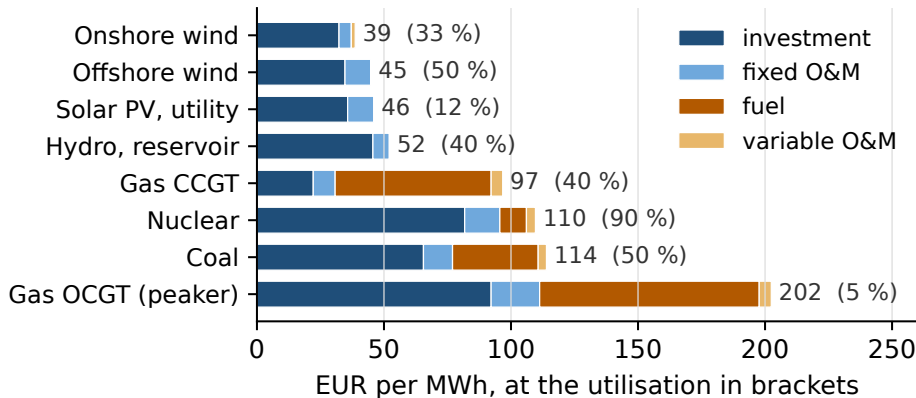
Which rows could balance a system with a lot of wind?

Learning curves



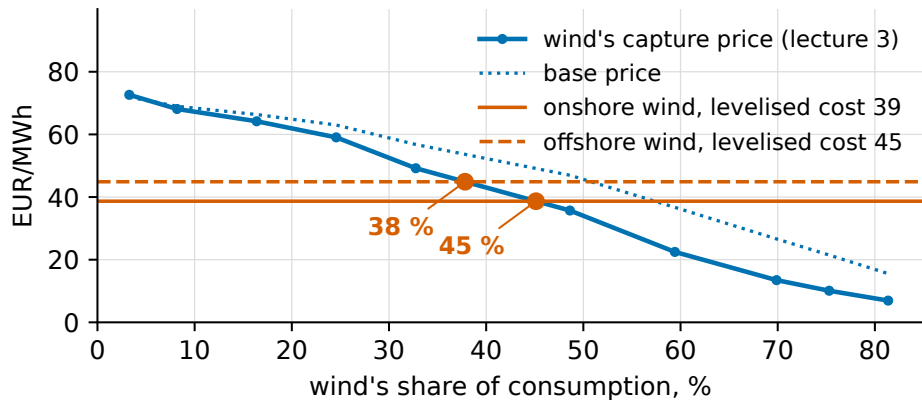
Our World in Data, CC BY 4.0: (a) IRENA, new projects worldwide, 2025 USD; (b) Way (2026), 2024 USD. Log scales.

By levelised cost, wind and solar have already won



Generation Technologies, Fig. 3.1: 2025 costs, the electricity rows.

Cannibalisation of revenue



Lecture 3's model, the fleet held fixed; levelised costs from *Generation Technologies*.

The hourly balance, and what can move

Every hour, in *Energy System Models'* notation:

$$\underbrace{\gamma_h \bar{q}_w}_{\text{wind and solar}} + \underbrace{q_{g,h}}_{\text{dispatchable plants}} + \underbrace{g_h^d - g_h^c}_{\text{storage}} = \underbrace{D_h}_{\text{demand}} + \underbrace{x_h}_{\text{net exports}}$$

- The first term moves with the weather. When it moves, some **blue** term must move with it, in the same hour.
- **Until recently:** demand was fixed, thermal plants did all the moving, and trade was at the margin.
- **As wind and solar replace thermal plants,** the terms that used to move are the ones being retired.

Four ways to balance

- **Flexible backup** ($q_{g,h}$): dispatchable plants raise their output when the wind drops, and lower it when the wind returns.
- **Flexible demand** (D_h): consumers move their use into the hours when power is plentiful.
- **Storage** ($g_h^d - g_h^c$): a surplus MWh is kept for a later hour, and some of it is lost on the way.
- **Trade** (x_h): a surplus is sold to the neighbours, and a shortfall is bought from them.

Today: storage, and its partner **curtailment**: throwing part of a surplus away.

A still, cold fortnight in January. Which option carries the system?

1. a large fleet of four-hour batteries
2. flexible demand: heat pumps and electric cars shift their use
3. backup plants and imports
4. curtailment

Storage technologies

A storage is two capacities

- **Power**, MW: how fast it charges and discharges. For a battery, the inverter.
- **Energy**, MWh: how much it holds. For a battery, the cells.
- **Duration** = MWh / MW: a 4-hour battery empties in four hours at full power.
- **Round-trip efficiency**: the share of a MWh put in that comes back out.

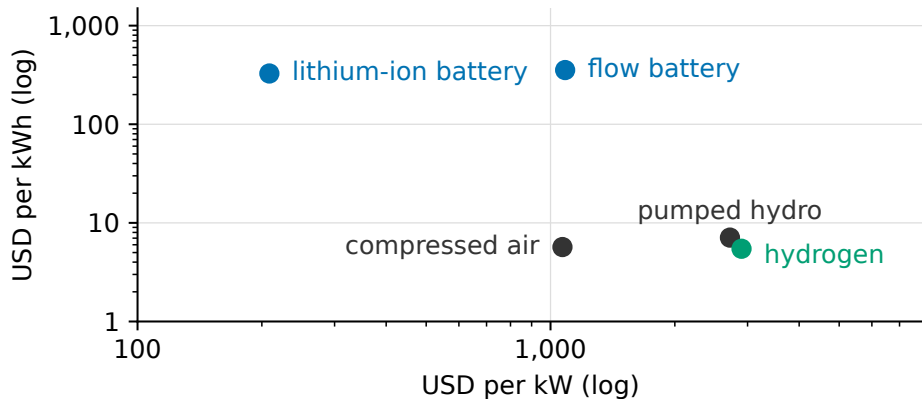
Energy System Models' battery

	2025	2050
inverter, EUR/kW	147	64
cells, EUR/kWh	128	80
4-hour system, EUR/kWh	165	96
round trip, inverter only	92%	
round trip, measured	82–83%	

The technology note's costs, the 2025 system capped at BNEF's turnkey price. Measured: the US battery fleet in 2019 (EIA) and PNNL's 2024 database, grid to grid.

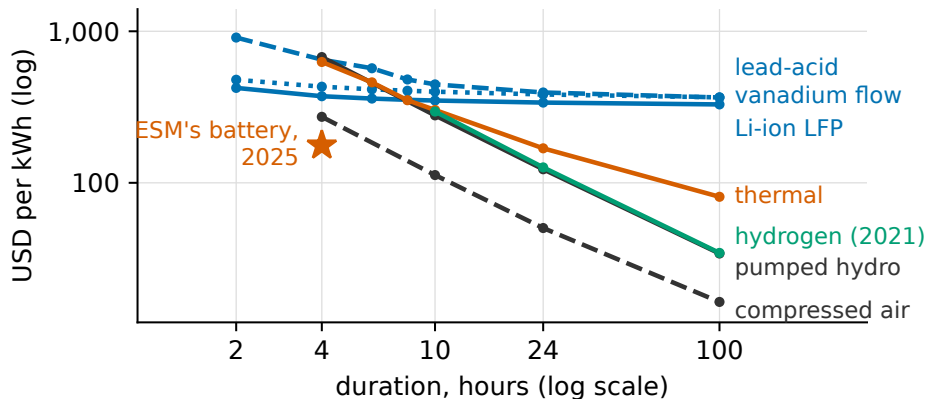
The two are bought separately, so they are sized separately. Part 2 asks which hours pay for each.

Every store is a price per MW and a price per MWh



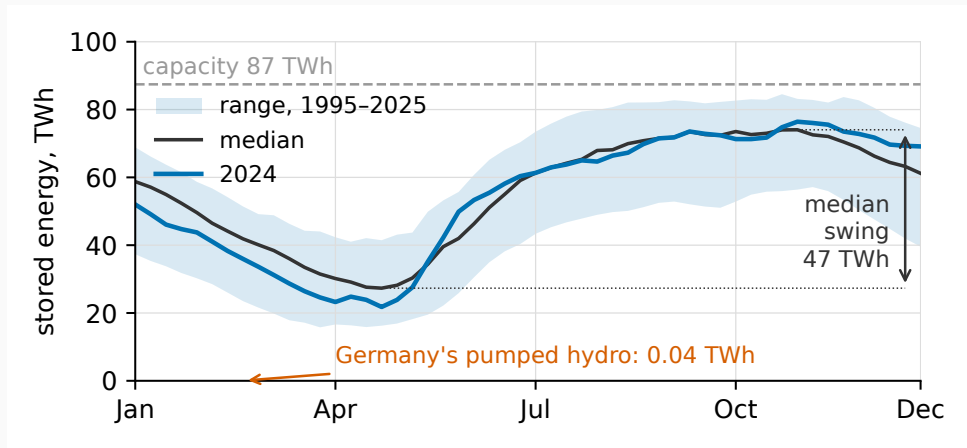
Reservoir hydro is off the chart: its tank is a lake, built long ago and filled by the weather. PNNL's storage database v2024, 100 MW projects, 2023 USD (hydrogen 2021).

Which store is cheapest depends on the duration



PNNL's storage database v2024, 100 MW, 2023 USD (hydrogen 2021). Star: ESM's 4-hour battery, 2025 European turnkey price.

Norway's reservoirs: a seasonal store at scale



Stored energy in Norway's reservoirs, week by week: NVE, reservoir statistics, the whole country, 1995-2025.

Hydro: cheap to own, and nearly built out

Cost-effective

- New reservoir hydro costs 52 EUR/MWh, against 39 for onshore wind. Almost all of it is the investment.
- Once built, it runs for 80 years, burns no fuel and costs 6.5 EUR/MWh to keep.
- The seasonal store comes with it: the tank is the lake, filled by rain and snow.

Further potential

- **Norway:** of 216 TWh a year that could be built, 64% is built and 23% protected. 23 TWh is left, 7.6 of it in upgrades.
- **Europe:** about half the technical potential is built, the highest share of any continent, best sites first.
- **What grows:** upgrades, and pumps between existing lakes. More MW, few new TWh.

Europe can add machines to the lakes it has. It cannot build another Norway.

Generation Technologies (at 40% utilisation); NVE (2020); JRC (2025).

Storage technologies: time scales and costs

	time scale	machine USD/kW	tank USD/kWh	round trip	in the course
lithium-ion battery	hours	209	327	83%	today, part 3
flow battery	hours	1,085	355	65%	
pumped hydro	hours to days	2,719	7	80%	today's paper
compressed air	hours to days	1,069	6	55%	
hydrogen in caverns	days to seasons	2,901	6	31%	lecture 7
reservoir hydro	seasons	2,430	a lake	–	lecture 5

PNNL's storage database v2024, 2023 USD (hydrogen 2021); reservoir hydro: *Generation Technologies*, a new plant.

Cheap machine, costly tank: hours. **Costly machine, cheap tank:** days and longer.

Workshop: Zerrahn et al. (2018)

Energy System Models, section 4: the storage model and its Lagrangian (§4.1), the arbitrage band, the table of duals.

Zerrahn et al. (2018), *On the economics of electrical storage for variable renewable energy sources*. Sections 1–5; §6 for the discussion.

Background: *Generation Technologies*, the batteries and *storage and prosumers* subsections. Optional: Butters et al. (2025), on whether arbitrage revenue pays for a battery.

Ambec and Crampes (2021), on flexible demand, moved to a block of its own later in the course.

The question

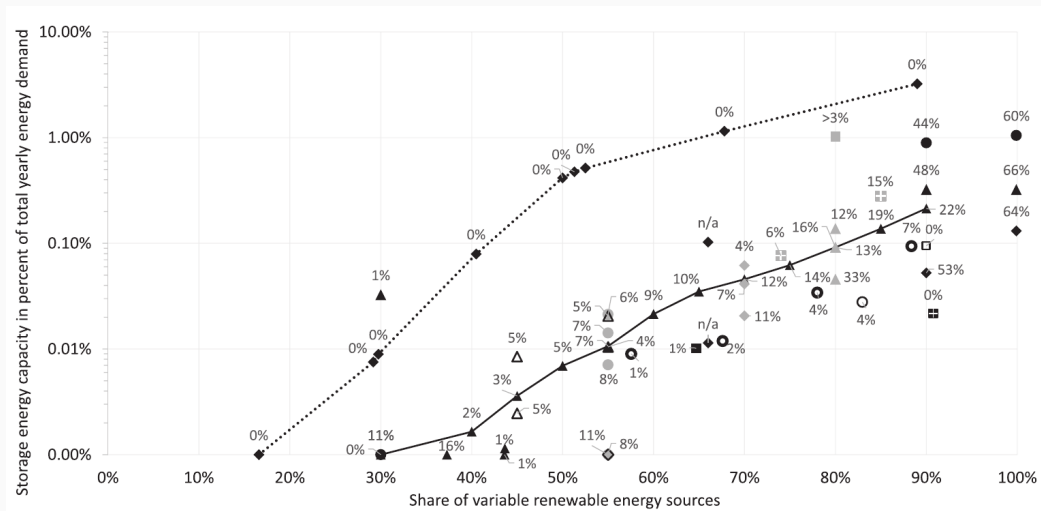
Sinn (2017): integrating wind and solar in Germany without waste needs a storage that takes every surplus MWh.

renewable share	50%	68%	89%
storage, GWh	2,100	5,800	16,300
share of annual demand	0.42%	1.15%	3.23%

Germany's pumped hydro holds just under 40 GWh.

Zerrahn et al. (2018): at 50%, a cost-minimising planner builds 35 GWh. **Which assumption produces a factor of sixty?**

The paper's Fig. 1: every study against Sinn



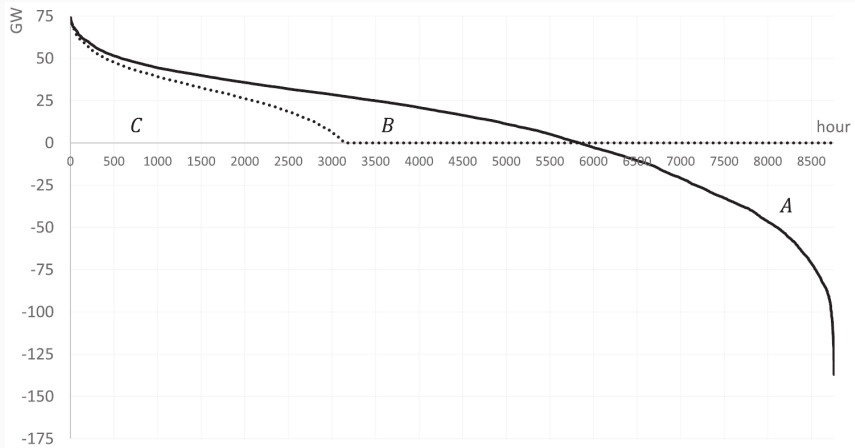
Part 1

Setup: the paper, figure by figure

- **Sinn's rule** (§3): every surplus MWh is stored and released at the first hour of positive residual load; the tank is the smallest that never runs dry.
- **The residual load duration curve** (§3.4): load minus wind and solar, for every hour of the year, sorted from highest to lowest. The surplus hours are below zero on the right.
- **Curtailment** (§4): two ways of throwing part of the surplus away, with Sinn's rule otherwise unchanged.
- Keep the paper open at its figures: each question reads one or two of Figs. 1–9.

The paper's Fig. 4 is on the next slide and on the worksheet.

Setup: the residual load duration curve



Zerrahn et al. (2018), Fig. 4, CC BY 4.0: Germany, 80% renewables, 2014. Solid: residual load; dotted: after storage use.

Why storage needs are two orders of magnitude below Sinn's

1. The claim: Sinn's claim, his two assumptions, Fig. 1.
2. Sinn's rule: the simulation, how the storage is used; realistic? (Figs. 2–3)
3. The residual load duration curve: A, B and C; how it treats time.
4. Power-oriented curtailment: Fig. 5, then Figs. 6–7.
5. Energy-oriented curtailment: Fig. 8, then Fig. 9 against Fig. 6.
6. *Perspective*: perfect foresight against Sinn's heuristic.

Stop point

The two assumptions; what passes through the tank and what sets it; at 50%, 5% curtailment: 0.3 TWh power-oriented, 19 GWh energy-oriented.

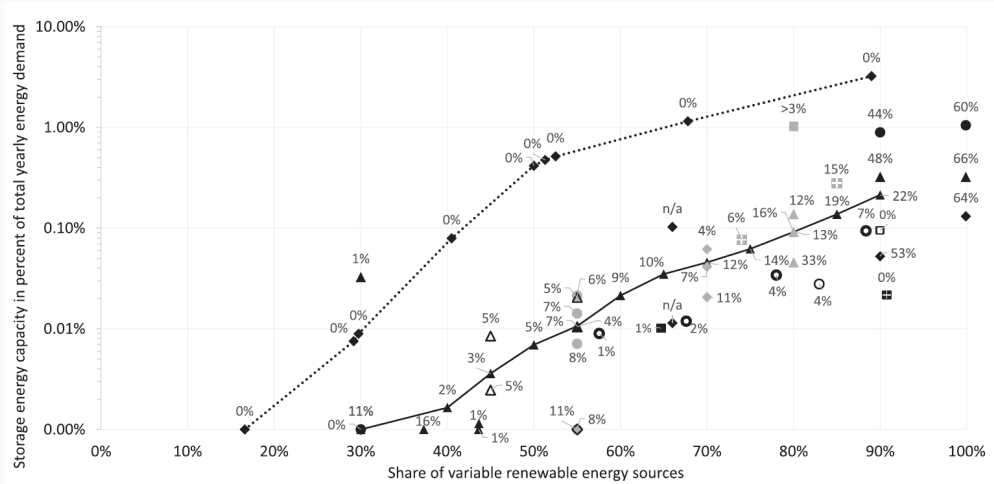
The full questions, the hints and the stretch are on the worksheet.

Solutions 1

Solutions 1, Q1–Q3: the questions

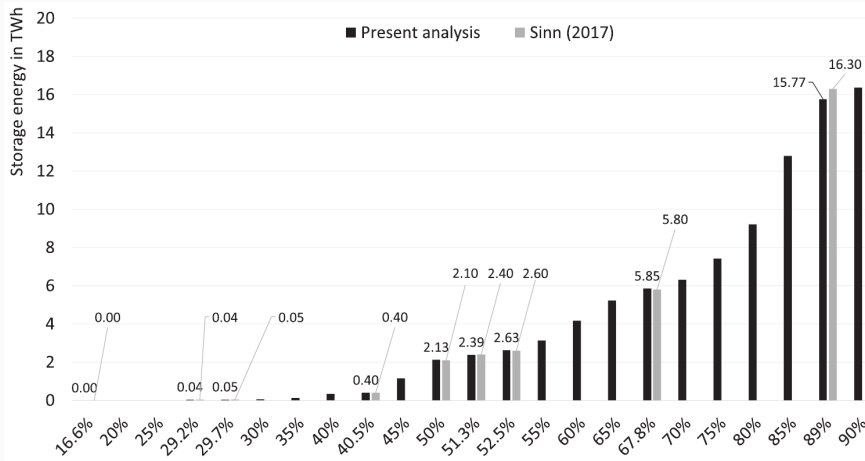
- Q1. What does Sinn claim, which two assumptions does he make, and what does Fig. 1 show?
- Q2. How does the paper simulate Sinn's rule, and how is the storage used? Do the results look realistic (Figs. 2–3)?
- Q3. What do the areas A, B and C of Fig. 4 measure? How does the residual load duration curve treat time?

Solutions 1, Q1: Sinn stores every surplus



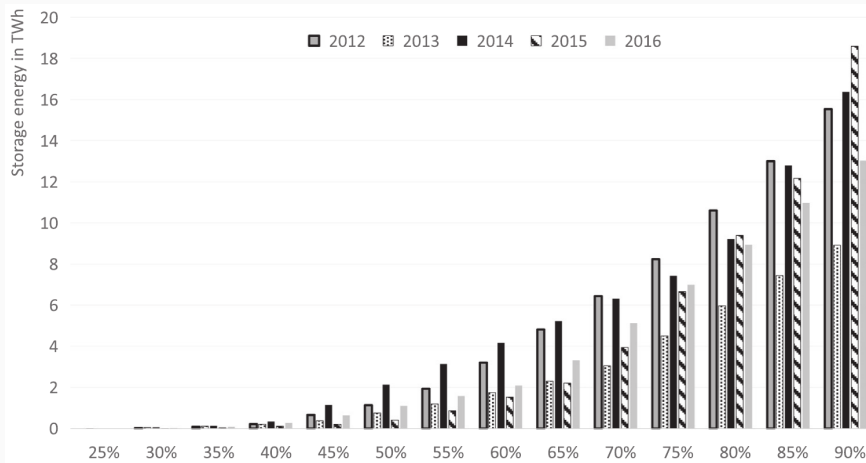
Zerrahn et al. (2018), Fig. 1, CC BY 4.0. Dotted: Sinn (2017). Labels: each study's curtailment, in percent.

Solutions 1, Q2: Sinn's rule, replicated



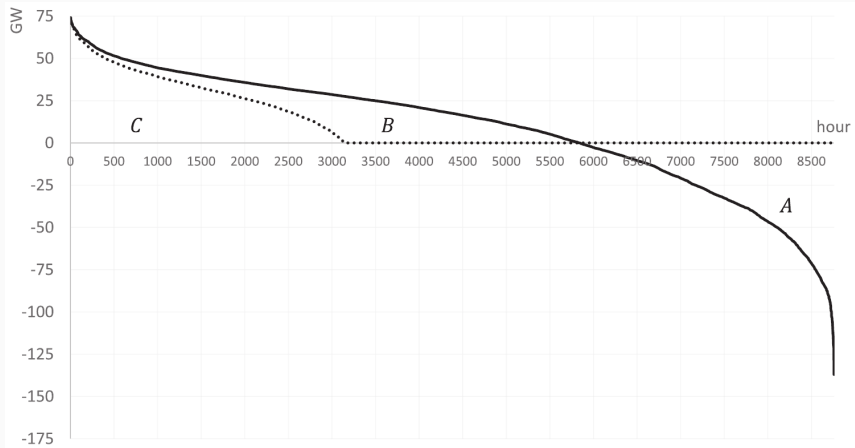
Zerrahn et al. (2018), Fig. 2, CC BY 4.0: storage energy by renewable share, Germany, 2014 data. Grey: Sinn (2017).

Solutions 1, Q2: one weather year



Zerrahn et al. (2018), Fig. 3, CC BY 4.0: Sinn's rule on five base years of German data.

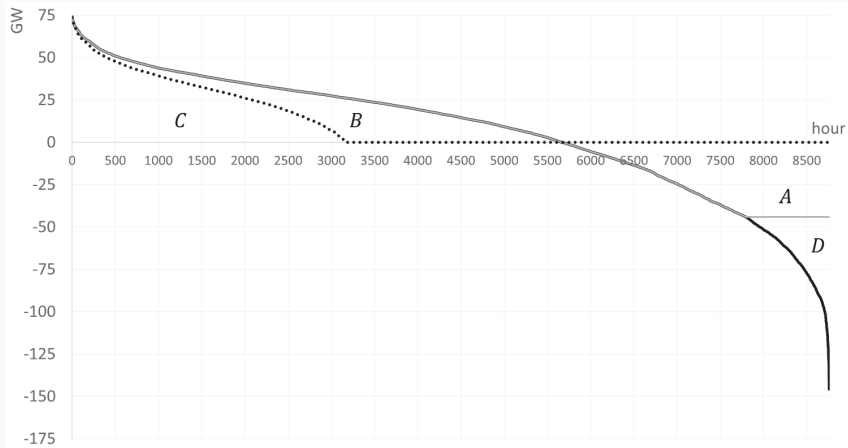
Solutions 1, Q3: the residual load duration curve has no clock



Zerrahn et al. (2018), Fig. 4, CC BY 4.0. A passes through the tank; the longest surplus run sets it, and sorting loses it.

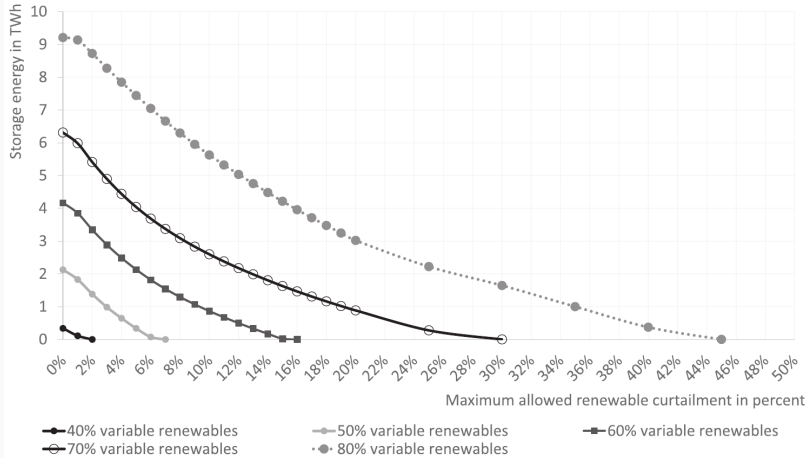
- Q4. Power-oriented curtailment: the idea on Fig. 5. What do Figs. 6 and 7 say about storage and curtailment?
- Q5. Energy-oriented curtailment: the idea on Fig. 8. Fig. 9 against Fig. 6.

Solutions 1, Q4: curtail above a threshold



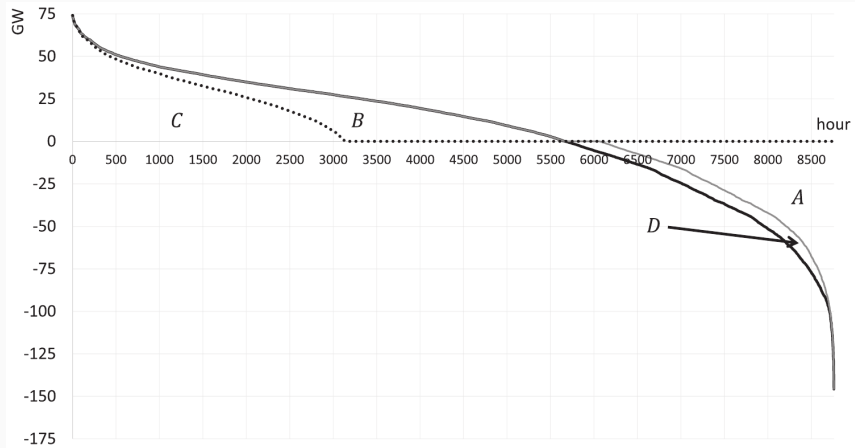
Zerrahn et al. (2018), Fig. 5, CC BY 4.0: 80% renewables. Grey: after curtailment above 44.1 GW (area D); dotted: and storage.

Solutions 1, Q4: storage and curtailment are substitutes



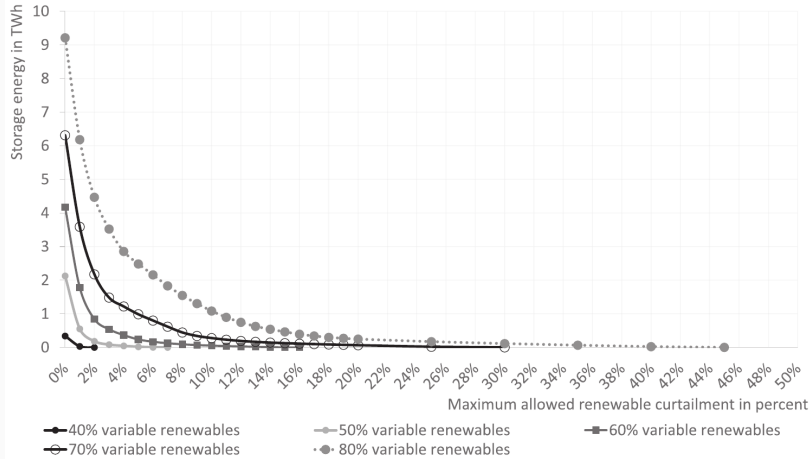
Zerrahn et al. (2018), Fig. 6, CC BY 4.0: storage energy against the curtailment budget, power-oriented.

Solutions 1, Q5: curtail when the tank is full



Zerrahn et al. (2018), Fig. 8, CC BY 4.0: 80%, the same 5% curtailed as in Fig. 5 (D). Grey: after curtailment; dotted: and storage.

Solutions 1, Q5: two orders of magnitude



Zerrahn et al. (2018), Fig. 9, CC BY 4.0: storage energy against the curtailment budget, energy-oriented. Same axes as Fig. 6.

Q6. *Perspective:* ESM's storage has perfect foresight; Sinn's releases at the first positive residual load. Into which hours does each release? What does foresight buy, and where is market reality (footnote 29)?

- **ESM's storage** releases where θ_h is beaten by the price: the far left of the residual load duration curve, the scarcest hours.
- **Sinn's** releases at the first positive residual load: somewhere in the middle. B looks like a band on top only because footnote 16 re-sorts the hours.
- **Foresight buys** the two values of \$5.2: displacing the most costly fuel, and shaving the peak a plant would otherwise be built for.
- **Market reality** “likely lies between the myopic and the perfect foresight cases” (footnote 29).

Part 2

Setup: the planner of §5

$$\min \underbrace{i^{se} N^{se} + i^{sp} N^{sp}}_{\text{storage: MWh and MW}} + \underbrace{\sum_c i^c N^c + i^r N^r}_{\text{plant and renewables}} + \underbrace{\sum_{c,h} v^c G_h^c + \sum_h v^s (S_h^{in} + S_h^{out})}_{\text{variable costs}}$$

(1) total cost

$$\text{s.t. } d_h = \sum_c G_h^c + G_h^r + S_h^{out}$$

(2) hourly balance (λ_h)

$$0 \leq G_h^c \leq N^c$$

(3a) plant capacity $(\underline{\mu}_{c,h}, \bar{\mu}_{c,h})$

$$\gamma_h N^r = G_h^r + S_h^{in} + C_h$$

(3b) renewable energy (ν_h)

$$G_h^r \geq 0, \quad C_h \geq 0$$

renewable use $(\underline{\mu}_{r,h}, \underline{\omega}_h)$

$$S_h = S_{h-1} + \eta_c S_h^{in} - S_h^{out} / \eta_d$$

(4a) storage balance (θ_h)

$$0 \leq S_h^{in} \leq N^{sp}$$

(4b) charging rating $(\underline{\phi}_h^c, \bar{\phi}_h^c)$

$$0 \leq S_h^{out} \leq N^{sp}$$

(4c) discharging rating $(\underline{\phi}_h^d, \bar{\phi}_h^d)$

$$0 \leq S_h \leq N^{se}$$

(4d) energy capacity $(\underline{\psi}_h, \bar{\psi}_h)$

$$\sum_{c,h} G_h^c \leq (1 - \alpha) \sum_h d_h$$

(5) renewable share (σ)

$$N^{se}, N^{sp}, N^c, N^r \geq 0$$

capacities $(\underline{\beta}^{se}, \underline{\beta}^{sp}, \underline{\beta}^c, \underline{\beta}^r)$

Setup: the Lagrangian, balances

$$\begin{aligned}
 \mathcal{L} = & \underbrace{- \left(i^{se} N^{se} + i^{sp} N^{sp} + \sum_c i^c N^c + i^r N^r + \sum_{c,h} v^c G_h^c + \sum_h v^s (S_h^{in} + S_h^{out}) \right)}_{\text{minus the total cost (1)}} \\
 & + \underbrace{\sum_h \lambda_h \left(\sum_c G_h^c + G_h^r + S_h^{out} - d_h \right)}_{\text{(2) hourly balance}} + \underbrace{\sum_{c,h} \left(\bar{\mu}_{c,h} (N^c - G_h^c) + \underline{\mu}_{c,h} G_h^c \right)}_{\text{(3a) plant capacity}} \\
 & + \underbrace{\sum_h \nu_h \left(\gamma_h N^r - G_h^r - S_h^{in} - C_h \right)}_{\text{(3b) renewable energy}} + \underbrace{\sum_h \left(\underline{\mu}_{r,h} G_h^r + \underline{\omega}_h C_h \right)}_{\text{renewable use}} + \dots
 \end{aligned}$$

As in ESM §4.1: the planner maximises minus the cost; each constraint is a slack that is nonnegative when it holds, times its multiplier; a lower bound $x \geq 0$ enters as $\underline{\mu} x$.

Setup: the Lagrangian, storage and target

$$\begin{aligned}
 \dots &+ \underbrace{\sum_h \theta_h (S_{h-1} + \eta_c S_h^{in} - S_h^{out}/\eta_d - S_h)}_{(4a) \text{ storage balance}} + \underbrace{\sum_h \left(\bar{\phi}_h^c (N^{sp} - S_h^{in}) + \underline{\phi}_h^c S_h^{in} \right)}_{(4b) \text{ charging rating}} \\
 &+ \underbrace{\sum_h \left(\bar{\phi}_h^d (N^{sp} - S_h^{out}) + \underline{\phi}_h^d S_h^{out} \right)}_{(4c) \text{ discharging rating}} + \underbrace{\sum_h \left(\bar{\psi}_h (N^{se} - S_h) + \underline{\psi}_h S_h \right)}_{(4d) \text{ energy capacity}} \\
 &+ \underbrace{\sigma \left((1 - \alpha) \sum_h d_h - \sum_{c,h} G_h^c \right)}_{(5) \text{ renewable share}} + \underbrace{\underline{\beta}^{se} N^{se} + \underline{\beta}^{sp} N^{sp} + \sum_c \underline{\beta}^c N^c + \underline{\beta}^r N^r}_{\text{capacities}}
 \end{aligned}$$

Free: $\lambda_h, \nu_h, \theta_h$, on the equalities. All others nonnegative, and zero unless their bound binds.

Setup: the first-order conditions

Capacities: minus the annuity, the multipliers summed over the year, and the lower bound

$$N^{sp} : -i^{sp} + \sum_h \left(\underbrace{\bar{\phi}_h^c}_{(4b)} + \underbrace{\bar{\phi}_h^d}_{(4c)} \right) + \underline{\beta}^{sp} = 0$$

$$N^{se} : -i^{se} + \sum_h \underbrace{\bar{\psi}_h}_{(4d)} + \underline{\beta}^{se} = 0$$

$$N^c : -i^c + \sum_h \underbrace{\bar{\mu}_{c,h}}_{(3a)} + \underline{\beta}^c = 0$$

$$N^r : -i^r + \sum_h \gamma_h \underbrace{\nu_h}_{(3b)} + \underline{\beta}^r = 0$$

Generation

$$G_h^c : -v^c + \lambda_h - \underbrace{\sigma}_{(5)} - \underbrace{\bar{\mu}_{c,h} + \underline{\mu}_{c,h}}_{(3a)} = 0$$

$$G_h^r : \lambda_h - \nu_h + \underline{\mu}_{r,h} = 0$$

$$C_h : -\nu_h + \underline{\omega}_h = 0$$

Storage operation

$$S_h^{in} : -v^s - \nu_h + \eta_c \theta_h - \underbrace{\bar{\phi}_h^c + \underline{\phi}_h^c}_{(4b)} = 0 \quad S_h^{out} : -v^s + \lambda_h - \theta_h / \eta_d - \underbrace{\bar{\phi}_h^d + \underline{\phi}_h^d}_{(4c)} = 0 \quad S_h : -\theta_h + \theta_{h+1} - \underbrace{\bar{\psi}_h + \underline{\psi}_h}_{(4d)} = 0$$

Complementary slackness on every bound. Your job in part 2 is to say what the conditions mean.

The cost-minimising model

1. The problem and its multipliers; the capacity conditions and σ ; why the tank stops far below Sinn's; *perspective*: ESM's storage model.
2. Fig. 10: D_1 , D_2 , B_1 – B_3 ; the two values of storage.
3. Germany's numbers at 50 and 90%; why duration and curtailment rise with the share.
4. Power-to-x: what a flexible new load does to storage.

Stop point

Which hours pay for pumps, basin and turbines, and what σ is. Germany: 35 GWh at 50%, no long-term storage, 35 $\rightarrow$ 4 GWh with power-to-x.

The full questions, the hints and the stretch are on the worksheet.

Solutions 2

- Q1. The problem and its multipliers. The four capacity conditions and the condition for G_h^c in words; why the planner builds turbines whose output is sometimes worth nothing. Why the tank stops far below Sinn's. *Perspective*: against ESM's storage model.

Solutions 2, Q1: which hours pay for what

a unit of	condition	is paid for by
pumps, N^{sp}	$i^{sp} = \sum_h (\bar{\phi}_h^c + \bar{\phi}_h^d) + \underline{\beta}^{sp}$	the hours at full power
basin, N^{se}	$i^{se} = \sum_h \bar{\psi}_h + \underline{\beta}^{se}$	the hours the tank is full
turbines, N^r	$i^r = \sum_h \gamma_h \nu_h + \underline{\beta}^r$	the hours its output is not curtailed
plant, N^c	$i^c = \sum_h \bar{\mu}_{c,h} + \underline{\beta}^c$	the hours it runs flat out
conventional MWh	$\lambda_h = v^c + \sigma + \bar{\mu}_{c,h} - \underline{\mu}_{c,h}$	σ is a tax on it

$\underline{\beta} > 0$ only for a capacity that is not built, $\underline{\mu}_{c,h} > 0$ only in hours a plant is off.

- **Q1. Curtailed hours:** $\nu_h = 0$, the spilled MWh was worth nothing; the turbine is paid by its other hours.
- **Q1. Perspective:** the operating conditions are ESM's arbitrage band; what is new is the investment layer.

Solutions 2, Q1: θ_h , the value of stored energy

θ_h is the value to the system of one more MWh in the tank at the end of hour h , in EUR/MWh: the shadow price of stored energy, ESM's *water value*. The condition for S_h says how it moves:

$$\theta_{h+1} = \theta_h + \overline{\psi}_h - \underline{\psi}_h.$$

- **Constant** while the tank is neither full nor empty: a MWh can be moved freely to the next hour, so its value cannot differ.
- **Jumps** up across an hour with a full tank, down across an hour with an empty one.
- **Pinned by prices:** when the storage charges below its rating, $\lambda_h = \eta_c \theta_h - v^s$; when it discharges below its rating, $\lambda_h = \theta_h / \eta_d + v^s$.

Solutions 2, Q1: the rating rents are price gaps

From the conditions for S_h^{in} and S_h^{out} , with $\nu_h = \lambda_h$ in a surplus hour:

$$\overline{\phi}_h^c - \underline{\phi}_h^c = \eta_c \theta_h - \lambda_h - v^s, \quad \overline{\phi}_h^d - \underline{\phi}_h^d = \lambda_h - \theta_h / \eta_d - v^s.$$

- **Charging flat out:** $\overline{\phi}_h^c > 0$ when stored energy is worth more than the price; largest when surplus is curtailed and $\lambda_h = 0$.
- **Discharging flat out:** $\overline{\phi}_h^d > 0$ when the price is above the value of the energy taken out of the tank.
- **So i^{sp} is paid by the gaps between the price and the value of stored energy, in the hours the rating binds.**

Solutions 2, Q1: the full-tank rent is a jump in θ_h

In an hour with a full tank, $\underline{\psi}_h = 0$, so $\bar{\psi}_h = \theta_{h+1} - \theta_h$. Over one fill, from the last hour the storage charges freely to the first it discharges freely:

$$\sum_{\text{full hours}} \bar{\psi}_h = \underbrace{\eta_d(\lambda^{out} - v^s)}_{\theta \text{ after}} - \underbrace{(\lambda^{in} + v^s)/\eta_c}_{\theta \text{ before}}.$$

- **The rent is a spread:** one more MWh of tank would take one more MWh at the fill price λ^{in} and release it at λ^{out} , net of losses.
- **So i^{se}** is paid by one spread for each time the tank is full, summed over the year.

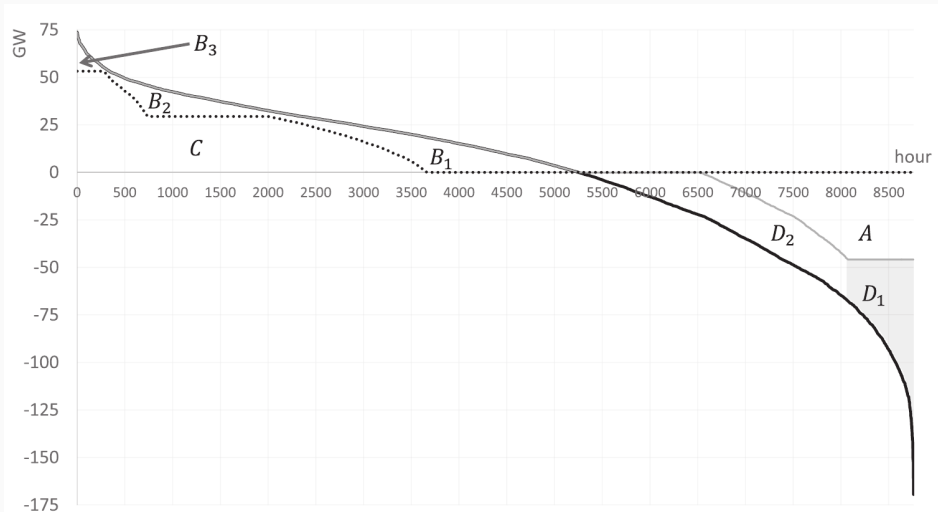
Solutions 2, Q1: why the tank stops far below Sinn's

1. The last MWh of basin is full only at the end of the **longest surplus runs**.
2. A skewed surplus has **few such runs**: a handful of full-tank hours a year, a handful of rents, far short of i^{se} .
3. So the planner stops building basin, and **curtails** the rest of the run, at no cost at the margin: those MWh have $\nu_h = 0$.

Sinn's tank is the one a planner builds if a MWh of basin is free, $i^{se} = 0$.

- Q2. In the paper's words, the areas D_1 , D_2 , B_1 – B_3 of Fig. 10; the two economic values of storage, and which areas carry each.
- Q3. Germany at 50% and 90%: energy, power, duration, and the conclusion on long-term storage. Why duration and curtailment rise.

Solutions 2, Q2: the paper's Fig. 10



- Q2. D_1 : curtailed above the 46 GW rating.
- Q2. D_2 : curtailed when the tank is full.
- Q2. B_1, B_2 : base and peak fuel displaced, the *arbitrage value*.
- Q2. B_3 : peak plant not built, the *capacity value*.

Solutions 2, Q3: Germany, and no long-term storage

renewable share	energy, GWh	power, GW	duration, h	curtailment
50%	35	6	6	5%
80%	~460			16%
90%	~1,080	58	19	~22%

The paper's §5.3; the 80 and 90% rows read off its Fig. 11.

- **Q3. The conclusion (p. 271):** “no need for a true long-term storage arises, that is, storing energy for weeks or months.”
- **Q3. Duration rises:** the MW is paid by the height of the spikes, the MWh by the length of the runs, and runs grow faster.
- **Q3. Curtailment rises:** the extra surplus comes in ever longer runs, and spilling it is cheaper than a tank to hold it.

Q4. Power-to-x (§5.4): what is added, and what happens to storage and curtailment at 50% and 90%? What does a shiftable load share with a storage, and what does it lack?

Solutions 2, Q4: a storage without a tank

- **Added:** 50 GW of perfectly flexible load, 2,000 full-load hours, 100 TWh a year, served by extra renewables.
- **At 50%:** storage 35 → **4 GWh**, curtailment 5% → under 1%. The new load eats the surplus.
- **At 90%:** storage unchanged. The load feeds on surplus that was being curtailed anyway, not on what the tank held.
- **The sentence:** a shiftable load consumes area A where it occurs. It is a storage without the tank, its cost and its losses, but it never gives the energy back.

Wrap-up

What carries forward

1. **Flexibility is the problem, not generation cost.** Wind and solar win on levelised cost and lose value past about 40%, unless some other term of the balance can move.
2. **A storage is two capacities, paid by different hours.** Power by the hours at full power, energy by the hours the tank is full; their price ratio sets the duration.
3. **Curtailement is storage's partner.** A skewed surplus fills the last MWh of tank too rarely to pay for it, so storage needs are two orders of magnitude below Sinn's and never seasonal.

Lecture 5 takes two more of the four ways to balance: flexible demand in the heat sector, and trade over wires.

Before lecture 5 (Monday 5 October): *Energy System Models*, sections 5 and 6; Ruhnau et al. (2020) and Gonzales et al. (2023). Full solutions to today's worksheet, part 3 included, go up after class.

- Ambec, Stefan and Claude Crampes (2021). **“Real-Time Electricity Pricing to Balance Green Energy Intermittency”**. *Energy Economics* 94, p. 105074.
- Butters, R. Andrew, Jackson Dorsey, and Gautam Gowrisankaran (2025). **“Soaking up the Sun: Battery Investment, Renewable Energy, and Market Equilibrium”**. *Econometrica* 93.3, pp. 891–927.
- Gonzales, Luis E., Koichiro Ito, and Mar Reguant (2023). **“The Investment Effects of Market Integration: Evidence from Renewable Energy Expansion in Chile”**. *Econometrica* 91.5, pp. 1659–1693.
- Ruhnau, Oliver, Lion Hirth, and Aaron Praktiknjo (2020). **“Heating with Wind: Economics of Heat Pumps and Variable Renewables”**. *Energy Economics* 92, p. 104967.
- Zerrahn, Alexander, Wolf-Peter Schill, and Claudia Kemfert (2018). **“On the Economics of Electrical Storage for Variable Renewable Energy Sources”**. *European Economic Review* 108, pp. 259–279.